



## PADRI VJEKO CENTRE

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## **L5-NIT | SERVER ADMINISTRATION | 10th-OCT-2025**

**ACADEMIC YEAR: 2025-2026**

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### **Question 1 - Multiple Choice / 10 Pts**

1. Which component allows multiple operating systems to run on a single physical server?
  - A. Server Manager
  - B. Active Directory
  - C. Hyper-V
  - D. PowerShell
2. Which of the following best defines a server core installation?
  - A. A minimal installation without GUI for performance and security
  - B. A full GUI-based installation
  - C. A client-side operating system
  - D. A web-based server setup
3. Which of these is NOT a web server software?
  - A. IIS
  - B. Apache
  - C. Hyper-V
  - D. Nginx
4. Which of the following is NOT a type of load balancing?
  - A. Layer-4
  - B. Layer-7
  - C. Layer-8
  - D. DNS-Based
5. Which command is used to list files and directories in Linux?
  - A. cd
  - B. ls
  - C. rm
  - D. mkdir
6. What is the default root directory symbol in Linux?
  - A. %
  - B. #
  - C. /
  - D. \*
7. Which of the following is a hardware requirement for installing Ubuntu Server?
  - A. At least 2GB RAM
  - B. At least 1 GHz CPU and 2GB RAM
  - C. 256MB RAM
  - D. None
8. What is the purpose of BIOS settings before Linux installation?
  - A. To set the correct boot order for installation media
  - B. To install software
  - C. To manage permissions

- D. To delete partitions
9. Which type of software is required to run a Linux server?
- A. Operating system and server applications
  - B. Office Suite
  - C. Game engine
  - D. Media Player
10. What does the 'r' permission mean in Linux?
- A. Read
  - B. Write
  - C. Execute
  - D. Remove

## Question 2 - True or False / 10 Pts

1. The Linux kernel is responsible for managing hardware resources.
2. Ubuntu Server comes with a graphical desktop interface by default.
3. Disk partitioning allows the separation of system and user files.
4. In Linux, the root user has limited administrative privileges.
5. The /home directory contains user files and personal folders.
6. The command rm is used to rename files.
7. Permissions in Linux are represented by symbols such as r, w, and x.
8. The sudo command allows normal users to perform administrative tasks.
9. Creating a bootable USB drive is part of post-installation verification.
10. Networking requirements are optional during server installation.

## Open Questions (30 Marks)

1. Compare the roles of Kernel and Shell in the Linux operating system. **(2 Marks)**
2. Explain how file permissions (r, w, x) help maintain security in a multi-user environment. **(2 Marks)**
3. Design a simple directory structure for a small organization with three departments: sales, hr, and it, showing subdirectories for each. **(3 Marks)**
4. Develop a short bash script that creates three directories (/data, /backup, /logs) and sets read/write permissions for the owner only. **(3 Marks)**
5. A user reports that the ls command isn't displaying hidden files. Identify the likely cause and suggest the correct command. **(2 Marks)**
6. Analyze how enabling FTP logging helps administrators. **(2 Marks)**
7. Describe the role of CNAME and PTR records in DNS management. **(2 Marks)**
8. Analyze how improper configuration of DNS zones affects domain name resolution. **(2 Marks)**
9. Evaluate DHCP vs. static IP addressing in a large enterprise with mobile devices. **(3 Marks)**
10. Explain why defining a proper IP range (scope) is critical for DHCP efficiency. **(2 Marks)**
11. Distinguish between Server Core and Desktop Experience installation types in terms of management and performance. **(2 Marks)**
12. Using a table, differentiate server administration for Windows Server OS vs Linux Server OS. **(2 Marks)**
13. Arrange the steps to host a website on IIS: **(3 Marks)**
  - ☐ Install IIS
  - ☐ Add Website
  - ☐ Assign Site Name and Physical Path
  - ☐ Test Connectivity

**Good Luck!**